Seongwon Jang

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Technical Skills

Languages: Python, C, C++
Frameworks & Libraries: PyTorch, TensorFlow, vLLM, DeepSpeed, LangChain, OpenCV, wanDB, MCP
DevOps & Tools: Docker, CMake
Middleware: ROS2
Platforms: Linux
SCM: Git
Databases: PostgreSQL, MySQL

Experience

Research Intern — Jul. 2025 – Sep. 2025
Ubiquitous AI Lab (UAI Lab), UNIST (Advisor: Prof. Taesik Gong) — *Ulsan, Korea*

- Investigated parameter-efficient fine-tuning (PEFT) methods for Mixture-of-Experts (MoE) model
- Benchmarked LoRA, ESFT, and the proposed technique on **OLMoE-1B-7B-Instruct** across major NLP benchmarks to evaluate fine-tuning efficiency and task performance.

Contests

AI for Industry Challenge | *PyTorch, ROS2, CV, RL* — Apr. 2026 – Present

- Responsible for overall project structure design and codebase organization.

Structural Stability Physical Reasoning AI | *PyTorch, VideoMAE, Knowledge Distillation* — Mar. 2026 – Mar. 2026

- Identified and resolved flaws in the existing data augmentation pipeline
- Validated the effectiveness of video frame-based augmentation through ablation studies.
- Selected optimal hyperparameters (model architecture, learning rate, weight decay) via systematic sweep experiments.
- Extracted feature vectors from video data using a **VideoMAE** encoder and designed regularization functions to improve generalization.
- Selected a knowledge distillation model and designed corresponding regularization functions for model compression.
- **Ranked 66th out of 484 teams** (top 17.7%).

Major Projects

Advanced Q&A Chatbot System for Public Offices | *LLM, vLLM, DeepSpeed, LoRA* — Jan. 2026 – Mar. 2026

- Accelerated inference and training using **vLLM** and **DeepSpeed**.
- Containerized the system with Docker Compose and integrated MCP for improved accessibility.
- Constructed high-quality synthetic Q&A datasets using Chain-of-Thought (CoT) prompting and DeepEval Synthesizer to secure golden datasets for fine-tuning.
- Applied SFT with **LoRA** to improve instruction-following performance.
- Deployed trained adapters and tokenizers to the HuggingFace Hub.

RoboDine: AI-Based Robotic Restaurant Automation Platform | *ROS2, YOLO, OpenCV* — Apr. 2025 – Jun. 2025

- Enhanced robot arm grasping by refining 6-DoF pose estimation with **YOLO-v8-pose** (Dropout + Pose Loss tuning) and a 5-frame moving average to suppress noisy coordinate predictions; applied **solvePnP** for full 6-DoF pose recovery.
- Implemented a foreign object detection algorithm for the robot arm using **Hough Circle Transform** and **DBSCAN** clustering.
- Built an obstacle type classification and avoidance pipeline for the mobile robot using Mediapipe; developed a ROS2 package publishing obstacle type, LiDAR distance, and robot ID as topics.
- Practiced Agile development with Git, Jira, and Confluence in a team of 6.

Minor Projects

Practice Pintos — Apr. 2026 – Present

- Practicing Standford University's Pintos project

Robo – Household Companion | *Raspberry Pi 4, Gemini API* — Mar. 2025 – May. 2025

- Developed a human–robot interactive home companion on Raspberry Pi 4; integrated microphone, touch sensors, and **Gemini API** for user intent recognition and action execution.

One-Stage Object Detector from Scratch | *TensorFlow, Keras* — Mar. 2025 – Jun. 2025

- Implemented a One-Stage Detector model from scratch using **TensorFlow** and **PyTorch**
- Trained and evaluated on the Pascal VOC 2012 dataset.

AIBAK: AI-Based Children's Book Kiosk Service | *GPT-4o-mini, LangChain, ElevenLabs* — Sep. 2024 – Dec. 2024

- Implemented AI-driven story and title generation using **ChatGPT-4o-mini** and **LangChain**; added TTS narration via ElevenLabs API.
- Awarded **Excellence Prize** at the In-Jeju IoT Innovation Challenge.

Facial Expression Recognition Web App | *Residual Masking Network, Flask, MySQL* Apr. 2024 – May. 2024

- Led a team to build a web-based facial expression recognition app using **Residual Masking Network**; developed the **Flask** + **MySQL** backend and managed the schedule via WBS.

Education

Sejong University Seoul, South Korea

B.S. in Electronics and Information Engineering *Mar. 2019 – Feb. 2026*

- GPA: 4.02 / 4.5 (Major GPA: 4.13 / 4.5), **Magna Cum Laude**

Honors & Awards

2025 – Grand Prize, ROS2 & AI Autonomous Driving / Robot Arm Developer Bootcamp

2024 – Excellence Prize, 1st In-Jeju IoT Innovation Challenge

2024 – Encouragement Award, Sejong University Reading Contest (May 2024, Dec. 2023); Grand Prize (Oct. 2023)

Certificates

2025 – Engineer Information Processing (Dec. 2025)

English

2026 – OPIc IH

2025 – TOEIC 925

Tutor Experience

C Language Algorithm Mentor Jan. 2024 – Dec. 2024

- Designed and authored algorithm problems in C, covering core data structures such as stacks, queues, and linked lists.
- Provided code reviews and optimized solutions for students, focusing on time complexity and memory efficiency.

Algorithm

Problem Solving Jun. 2025 – Present

- Achieved **Gold 1** rank on Baekjoon Online Judge (solved over 430+ problems).